👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.DS.3

Shape of Data Distributions

Lesson 8-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes `Level 3 Enrichment`

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can describe the shape of a data distribution as symmetric, skewed, or having clusters and gaps.

Language Objective: I can explain my description using the words symmetric, skewed, cluster, and gap.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Symmetric | **Skewed** | **Cluster** | **Gap** | **Data distribution** | **Variability**

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A distribution where both sides look like mirror images is ______________.

2. A distribution with a tail stretching out to one side is ______________.

3. A group of data values bunched close together is a ______________.

4. An interval where there are no data values is a ______________.

5. The way data values are spread out across a range is the ______________.

6. How much the values in a data set differ from one another is the ______________.

Reflect — Exit Ticket

A data set has most values clustered between 40–60, with a few values at 90–100. What is the shape of the distribution and which measure of center is best?

A. Skewed right; use median

B. Symmetric; use mean

C. Skewed left; use median

D. Uniform; use mean

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Symmetric
2. **Notes 2:** Skewed
3. **Notes 3:** Cluster
4. **Notes 4:** Gap
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Shape of Data Distributions is skipping the key idea: "Symmetric data mirrors around the center; skewed data has a long tail, and the skew is named for the direction the tail points." — always check your work against this rule before you submit.*
8. **Try It 1:** A. Symmetric with one cluster around the peak — *One peak in the middle with roughly even sides means the data mirrors around the center, so it is symmetric with a single cluster at the peak.*
9. **Try It 2:** A. A cluster (70-85%), a gap (50-65%), and a left tail — so skewed left — *The data clusters at the high end (70-85%), there is an empty gap (50-65%), and the few low values form a tail on the left, so the display is skewed left.*
10. **Exit Ticket:** A. Skewed right; use median — *The data clusters on the left (40–60) with a tail to the right (90–100). This is skewed right. The median is best because the mean would be pulled toward the high values.*